MODEL LAB · 2026-10-28

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)	11	17,800	2.92%	0.5998
served prior	priors compared	held-out plate appearances	held-out HR rate	best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-13, scored on the 17,800 that came after (actual HR rate 2.92%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5813	1.62×	0.02826	0.13096	0.00454
KNN comps Prior is the pooled rate of the nearest comparables.	0.5998	1.64×	0.02821	0.12987	0.00605
SVR Support-vector regression on the same feature matrix.	0.5925	1.52×	0.02826	0.13049	0.00693
Random forest Forest regression; a learned similarity metric.	0.5951	1.50×	0.02825	0.13018	0.00678
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5998	1.64×	0.02821	0.12987	0.00605
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5998	1.64×	0.02821	0.12987	0.00605
Linear regression Ordinary least squares on the same features.	0.5964	1.58×	0.02825	0.13026	0.00487
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5953	1.60×	0.02830	0.13055	0.00609
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5955	1.56×	0.02824	0.13016	0.00503
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5945	1.45×	0.02829	0.13055	0.00733

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **260** knn **260** svr **260** rf **260** core **260** ensemble **260** linear **260** logistic **260** nn **260** xgb **260**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,620 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.6456	1.58×	0.02291	0.11126	0.01790
KNN comps Prior is the pooled rate of the nearest comparables.	0.6062	1.58×	0.02288	0.10980	0.01007
SVR Support-vector regression on the same feature matrix.	0.5783	1.58×	0.02323	0.11613	0.02309
Random forest Forest regression; a learned similarity metric.	0.6000	1.32×	0.02320	0.11319	0.02260
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6062	1.58×	0.02288	0.10980	0.01007
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6062	1.58×	0.02288	0.10980	0.01007
Linear regression Ordinary least squares on the same features.	0.6112	1.32×	0.02322	0.11383	0.02025
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6015	1.32×	0.02354	0.11442	0.02384
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6056	1.58×	0.02300	0.11128	0.01557
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5934	1.05×	0.02361	0.11640	0.02762

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jackson Chourio 15.4%, Junior Caminero 14.9%, Jake Bauers 14.1%, William Contreras 13.8%, Jonathan Aranda 11.7%, Christian Yelich 11.6%
Flat league	5/6	Jackson Chourio 15.7%, Junior Caminero 14.2%, William Contreras 13.6%, Jake Bauers 13.0%, Yandy Díaz 11.9%, Christian Yelich 11.7%
KNN comps	6/6	Jackson Chourio 16.0%, Junior Caminero 14.7%, Jake Bauers 14.2%, William Contreras 13.8%, Christian Yelich 11.9%, Jonathan Aranda 11.8%
SVR	5/6	Jackson Chourio 16.3%, Junior Caminero 16.0%, Jake Bauers 14.0%, William Contreras 13.7%, Yandy Díaz 12.1%, Jonathan Aranda 11.4%
Random forest	5/6	Jackson Chourio 16.2%, Junior Caminero 15.2%, Jake Bauers 14.2%, William Contreras 13.7%, Yandy Díaz 11.9%, Christian Yelich 11.7%
Core (KNN+SVR)	6/6	Jackson Chourio 16.0%, Junior Caminero 14.7%, Jake Bauers 14.2%, William Contreras 13.8%, Christian Yelich 11.9%, Jonathan Aranda 11.8%
Forest blend	6/6	Jackson Chourio 16.0%, Junior Caminero 14.7%, Jake Bauers 14.2%, William Contreras 13.8%, Christian Yelich 11.9%, Jonathan Aranda 11.8%
Linear regression	6/6	Jackson Chourio 16.2%, Junior Caminero 14.9%, Jake Bauers 14.0%, William Contreras 13.7%, Christian Yelich 11.8%, Jonathan Aranda 11.7%
Logistic regression	5/6	Jackson Chourio 16.2%, Junior Caminero 15.3%, Jake Bauers 14.0%, William Contreras 13.7%, Yandy Díaz 11.6%, Jonathan Aranda 11.5%
Neural net (torch)	5/6	Jackson Chourio 16.2%, Junior Caminero 15.3%, Jake Bauers 14.1%, William Contreras 13.8%, Yandy Díaz 11.7%, Jonathan Aranda 11.6%
XGBoost	5/6	Jackson Chourio 16.2%, Junior Caminero 15.5%, Jake Bauers 14.3%, William Contreras 13.7%, Yandy Díaz 12.0%, Christian Yelich 11.6%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Jackson Chourio**	MIL	**15.4%**	15.7%	16.0%	16.3%	16.2%	16.0%	16.0%	16.2%	16.2%	
2	**Junior Caminero**	TB	**14.9%**	14.2%	14.7%	16.0%	15.2%	14.7%	14.7%	14.9%	15.3%	
3	**Jake Bauers**	MIL	**14.1%**	13.0%	14.2%	14.0%	14.2%	14.2%	14.2%	14.0%	14.0%	
4	**William Contreras**	MIL	**13.8%**	13.6%	13.8%	13.7%	13.7%	13.8%	13.8%	13.7%	13.7%	
5	**Jonathan Aranda**	TB	**11.7%**	11.1%	11.8%	11.4%	11.4%	11.8%	11.8%	11.7%	11.5%	
6	**Christian Yelich**	MIL	**11.6%**	11.7%	11.9%	11.3%	11.7%	11.9%	11.9%	11.8%	11.4%	
7	**Yandy Díaz**	TB	**11.5%**	11.9%	11.6%	12.1%	11.9%	11.6%	11.6%	11.7%	11.6%	
8	**Brice Turang**	MIL	**10.4%**	10.6%	10.1%	10.3%	10.0%	10.1%	10.1%	9.7%	9.7%	
9	**Ryan Vilade**	TB	**9.5%**	8.9%	9.4%	9.1%	9.4%	9.4%	9.4%	9.1%	8.9%	
10	**Garrett Mitchell**	MIL	**9.5%**	9.1%	9.4%	8.8%	9.1%	9.4%	9.4%	9.1%	9.1%	
11	**Jonny DeLuca**	TB	**7.3%**	8.4%	7.4%	7.3%	7.0%	7.4%	7.4%	7.4%	7.7%	
12	**Cooper Pratt**	MIL	**7.1%**	8.5%	7.0%	6.5%	6.3%	7.0%	7.0%	6.0%	6.1%	
13	**Liam Hicks**	TB	**6.8%**	7.8%	6.8%	7.5%	7.4%	6.8%	6.8%	7.2%	7.1%	
14	**David Hamilton**	MIL	**6.5%**	7.7%	6.3%	6.2%	5.8%	6.3%	6.3%	6.4%	6.3%	
15	**Sal Frelick**	MIL	**6.5%**	8.1%	6.8%	6.1%	6.0%	6.8%	6.8%	6.5%	6.6%	
16	**Ben Williamson**	TB	**4.5%**	7.6%	5.1%	4.5%	4.5%	5.1%	5.1%	3.2%	4.9%	
17	**Taylor Walls**	TB	**3.7%**	4.8%	3.5%	2.6%	2.7%	3.5%	3.5%	3.4%	3.5%	
18	**Chandler Simpson**	TB	**2.0%**	4.5%	2.6%	1.2%	1.3%	2.6%	2.6%	1.1%	2.3%	

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Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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